

is the amount of time the processor spends on setting the system clock from the interrupt received from the hardware clock. Another problem is the jitter associated with preemptive task switching which sometimes leads to error in the accuracy of time.

There are many solutions for providing high resolution timer services, one of the easier one to implement is to write the current timestamp in the memory accessible by an application software. One such example is a Pentium processor, a user thread can be read the Pentium time stamp counter. This counter resets to 0 when the system restarts and increments at every processor cycle. Based on today's processor speeds this counter increments several times during a nanosecond. Mind well this is a platform specific implementation and it will be difficult to port the application which uses this facility to a platform which may have a different resolution clock or may not have this facility at all.

Along with the clock services the operating system also provides programmers with two types of timers - periodic timers and aperiodic(one shot) timers.

Periodic timers

Periodic timers are used to keep track of repetitive tasks like polling or performing some activity periodically. After a periodic timer is set, each time it expires a handler routine is called and the timer gets re-inserted into the timer queue.

Aperiodic (one shot) timers

These timers are set to expire only once. One popular example of an one-shot timer is the watchdog or the grenade timer. Watchdog timers are used extensively in real time systems, they form a part of the failsafe mechanism which is activated as soon as the timer expires. It is usually used to handle deadline misses and calling the exception handling routine when the timer expires.

Common RTOSes

We'll discuss some of the more prevalent operating systems available commercially in the markets today.

pSOS

Supported Platform: Motorola 68000

Kernel Size: 12K